# Heart Disease Analysis Using Tableau

## 1. INTRODUCTION

### 1.1 Project Overview

Heart disease is one of the leading causes of death worldwide and continues to increase due to unhealthy lifestyle habits, lack of physical activity, obesity, smoking, alcohol consumption, and chronic conditions such as diabetes and hypertension. Although healthcare systems generate a large amount of data, understanding this data in a meaningful way is often difficult without proper analytical tools.

The **Heart Disease Analysis** project focuses on analysing heart disease–related data and presenting insights through interactive visualisations. The project uses **Tableau** to transform raw datasets into dashboards and stories that clearly show trends, patterns, and relationships among various health factors such as age, gender, lifestyle habits, and medical conditions.

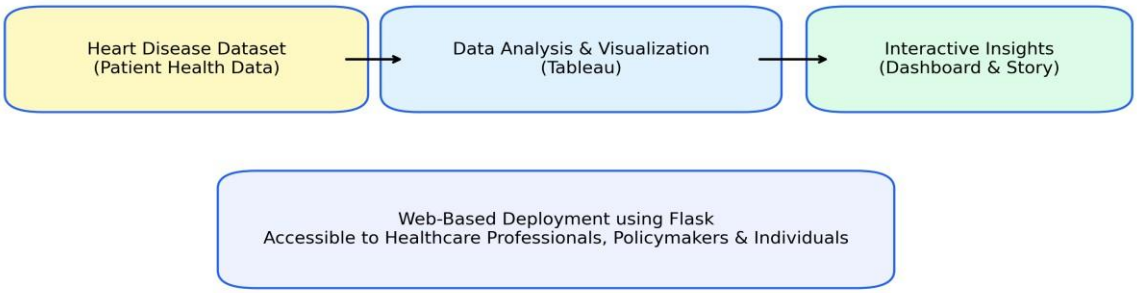



**Figure 4: Project Overview**

To improve accessibility, the dashboards and stories are integrated into a **Flask-based web application**, allowing users to explore insights easily through a browser. This approach helps healthcare professionals, policymakers, and individuals understand heart disease risks more effectively and supports informed decision-making.

### 1.2 Purpose

The purpose of the **Heart Disease Analysis** project is to make complex healthcare data simple, understandable, and actionable. Instead of using static reports or raw datasets, the project presents information through interactive dashboards that allow users to explore data dynamically.

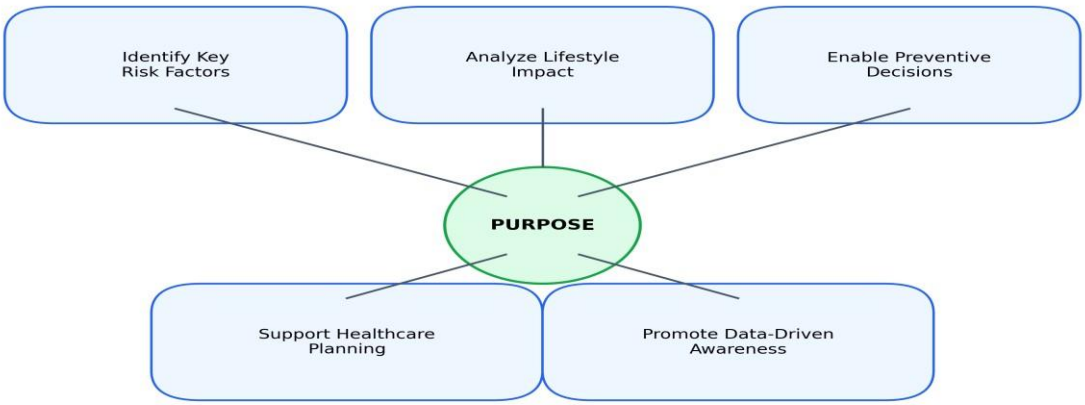



**Figure 5: Purpose of the Project**

The project aims to identify key risk factors associated with heart disease, analyse the impact of lifestyle choices such as smoking and physical activity, and support preventive healthcare decisions. It also helps healthcare professionals and policymakers plan better health strategies while encouraging individuals to adopt healthier lifestyles based on data-driven insights.

# 2. IDEATION PHASE

The ideation phase is a critical stage in the project development process, as it focuses on clearly understanding the problem domain, analysing user needs, and generating meaningful solution ideas. In the context of healthcare analytics, improper interpretation of medical data can lead to delayed decisions and ineffective preventive measures. Therefore, this phase emphasizes problem identification, user empathy, and creative thinking to design a solution that is practical, user-friendly, and impactful.

## 2.1 Problem Statement

Heart disease remains one of the major causes of mortality worldwide, largely due to lifestyle factors, lack of awareness, and delayed diagnosis. Healthcare organizations collect massive amounts of patient data related to heart disease, including demographic details, lifestyle habits, and medical conditions. However, this data is often stored in raw or tabular formats, making it difficult to analyse and interpret effectively.

Healthcare professionals, policymakers, and individuals face challenges in identifying meaningful patterns and correlations from this data. Traditional reporting methods and static analysis tools fail to provide interactive insights that can support preventive decision-making. As a result, potential risk factors and early warning signs may go unnoticed.

There is a need for an interactive, visual, and user-friendly system that can analyze heart disease data and present insights clearly to support awareness, early intervention, and preventive healthcare decisions.

## 2.2 Empathy Map Canvas

The empathy map canvas is used to gain a deeper understanding of the target users and their expectations from the system. The primary users of this project include healthcare professionals, public health policymakers, and individuals who are concerned about heart disease risks. Understanding their thoughts, emotions, challenges, and goals is essential for designing a solution that addresses real-world problems.

- **Think & Feel:** Users are concerned about increasing heart disease cases and want reliable insights to support better health decisions.
- **See:** Users are exposed to large datasets, complex reports, and statistical information that are difficult to interpret.
- **Hear:** Medical recommendations, public health guidelines, and awareness campaigns influence user perception and decisions.
- **Say & Do:** Users actively seek tools that simplify analysis and help them explore health data visually.
- **Pain:** Difficulty in understanding medical data, lack of clarity, and delayed identification of risk factors.
- **Gain:** Clear visual insights, faster understanding of trends, and improved preventive actions.

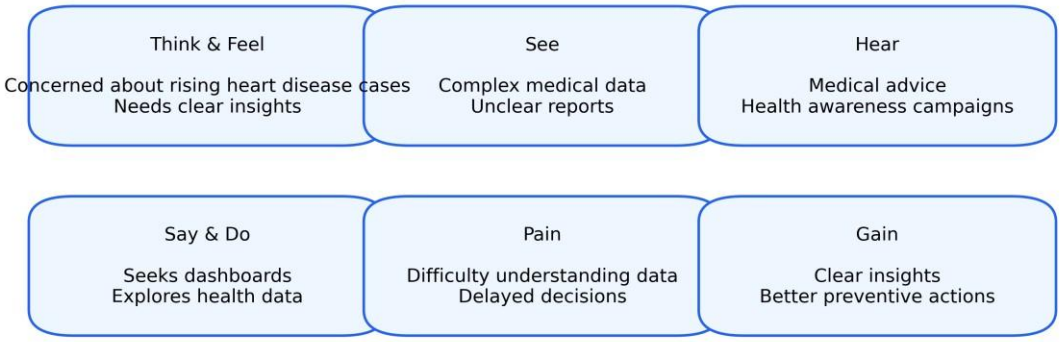



**Figure 9: Empathy Map Canvas**

### 2.3 Brainstorming

Brainstorming was carried out to explore multiple ideas and approaches that could effectively solve the identified problem. During this stage, different solution concepts were discussed, focusing on data clarity, usability, and accessibility. The objective was to identify a solution that could convert complex heart disease data into meaningful and actionable insights.

Several ideas emerged during the brainstorming process, including the use of interactive dashboards, visual storytelling, and web-based deployment. These ideas were evaluated based on feasibility, impact, and ease of use. The final solution approach was selected by prioritizing simplicity, interactivity, and real-time data exploration.

Key ideas generated during brainstorming include:

- Interactive dashboards for exploring heart disease trends
- Visual analysis of lifestyle factors such as smoking, alcohol consumption, and physical activity
- Identification of high-risk groups using comparative visualizations
- Story-based presentation of insights for better understanding
- Web-based access to ensure availability across different user groups

# 3. REQUIREMENT ANALYSIS

The Requirement Analysis phase plays a vital role in transforming the problem identified during the ideation phase into a clear and structured solution. This phase focuses on understanding how users interact with the system, defining what the system must accomplish, analysing how data flows within the system, and selecting appropriate technologies for implementation. A well-defined requirement analysis ensures that the proposed system is efficient, user-friendly, and aligned with project objectives.

## 3.1 Customer Journey Map

The customer journey map illustrates the complete interaction flow between the user and the Heart Disease Analysis system. It helps in understanding user behavior, expectations, and pain points at each stage of interaction. By mapping the journey, the system can be designed to provide a smooth and intuitive user experience.

The journey begins when a user accesses the web application through a browser. Upon entering the application, the user navigates to the dashboard or story section, where heart disease insights are presented visually. The user can then apply filters such as age group, gender, lifestyle habits, or health conditions to analyze specific patterns. Through interactive charts and dashboards, the user explores trends, correlations, and comparisons related to heart disease. Finally, the user interprets the insights to improve awareness, support healthcare decisions, or adopt preventive lifestyle measures.

This journey emphasizes the importance of simplicity, responsiveness, and interactivity, ensuring that even users without technical expertise can easily understand the insights presented.

## 3.2 Solution Requirement

Solution requirements define the expected behaviour and performance of the system. These requirements are broadly categorized into functional and non-functional requirements to ensure both correctness and quality of the solution.

### Functional Requirements

Functional requirements describe the core operations that the system must perform:

- The system should allow users to load and analyse heart disease datasets.
- The system should display interactive dashboards and story-based visualizations.
- Users should be able to filter, compare, and explore data across multiple parameters.
- The system should provide web-based access to insights through a user-friendly interface.

### Non-Functional Requirements

Non-functional requirements focus on the quality and performance aspects of the system:

- **Usability:** The system should be easy to navigate with minimal learning effort.

- **Performance:** Dashboards and stories should load efficiently and respond smoothly to user interactions.
- **Scalability:** The system should support future enhancements, such as larger datasets or additional health parameters.
- **Availability:** The system should be accessible anytime through a standard web browser.
- **Reliability:** The system should consistently display accurate and up-to-date visualizations.

**Figure 12: Solution Requirements**

### 3.3 Data Flow Diagram

The Data Flow Diagram (DFD) provides a graphical representation of how data moves through the system. It helps in understanding how user requests are processed and how raw data is transformed into meaningful visual insights.

In the Heart Disease Analysis system, the process begins when the user sends a request through the web interface. The application then accesses the heart disease dataset and processes the data using Tableau. Tableau performs the necessary data handling and visualization tasks, converting raw data into dashboards and stories. These visual outputs are then delivered back to the user through the web application, allowing interactive exploration.

The DFD ensures clarity in system operations and helps identify data sources, processing steps, and output destinations.

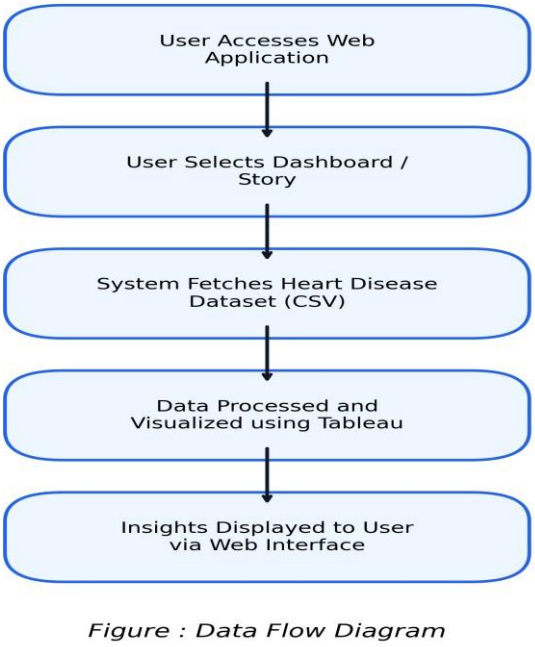



**Figure 13: Data Flow Diagram**

### 3.4 Technology Stack

The technology stack represents the collection of tools and technologies used to develop and deploy the system. Choosing an appropriate technology stack is essential for achieving performance, scalability, and maintainability.

- **Frontend Technologies:** HTML and CSS are used to design a clean and responsive user interface.
- **Backend Technologies:** Python with Flask is used to integrate the web application and manage routing.
- **Visualization Tool:** Tableau is used to create interactive dashboards and stories for data analysis.
- **Data Source:** The heart disease dataset is stored in CSV format for easy access and integration.
- **Deployment Environment:** The application is deployed in a web-based environment, making it accessible across devices.

This technology stack supports efficient development, interactive visualization, and seamless user experience, making it suitable for healthcare analytics applications.

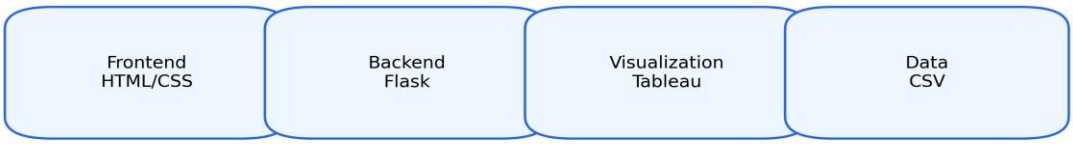



**Figure 14: Technology Stack**

# 4. PROJECT DESIGN

The project design phase defines how the identified problem is translated into a practical and efficient solution. This phase focuses on aligning the problem with an appropriate solution approach, describing the proposed system in detail, and explaining the overall system architecture. A strong project design ensures clarity, usability, scalability, and effective system performance.

## 4.1 Data Collection & Extraction from Database

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes and generate insights from the data.

### Downloading the dataset

Acquire the finalized dataset required for the Tableau project, ensuring it is clean, relevant, and aligned with the defined problem. Validate data integrity and readiness for analysis and visualization tasks.

Link: https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view

### Storing Data in DB & Perform SQL Operations

Store the collected dataset in a structured database system. Use SQL operations to clean, filter, transform, and prepare the data for seamless integration with Tableau for visualization and analysis.

### Connect DB with Tableau

Establish a secure and reliable connection between the database and Tableau. Ensure realtime or scheduled data access for creating interactive dashboards and performing dynamic data analysis within the Tableau environment.

## 4.2 Data Preparation

Clean, transform, and organize the connected data to ensure consistency and accuracy. Create calculated fields, handle null values, and structure the data appropriately for effective visualization and insightful analysis in Tableau.

### Prepare the Data for Visualization

In this step, we focus on preparing the dataset for visualization in Tableau. Fortunately, the dataset we're working with has already been pre-cleaned, meaning major cleaning steps such as handling missing values, removing duplicates, and correcting inconsistencies have already been taken care of.

However, even with a clean dataset, it's still essential to go through a brief review process to ensure it's truly ready for analysis:

- Data Review & Exploration
  While the dataset is clean, it’s good practice to explore it briefly—checking data types, value ranges, and distributions. This helps us understand the structure, identify any potential outliers, and gain familiarity with the data we’ll be visualizing.
- Filtering and Structuring for Purpose
  Depending on the business question, we may still need to filter the data to focus on specific subsets—such as certain time periods, regions, or product categories. Structuring the data to match the visualization goal helps ensure relevance and clarity.
- Field Renaming & Final Formatting
  To enhance clarity in Tableau, we ensure field names are intuitive and consistent. We also check for proper data types (e.g., date fields, numeric values) and relationships if the dataset spans multiple tables.
- Optional Calculated Fields
  If needed, we can create calculated fields (e.g., profit margin, growth rate) to support deeper analysis. Even with a clean dataset, these additions can make our visualizations more insightful.
- Validation for Accuracy
  Lastly, a quick validation against the source or summary metrics ensures everything is accurate. This final step helps maintain trust in the insights generated.

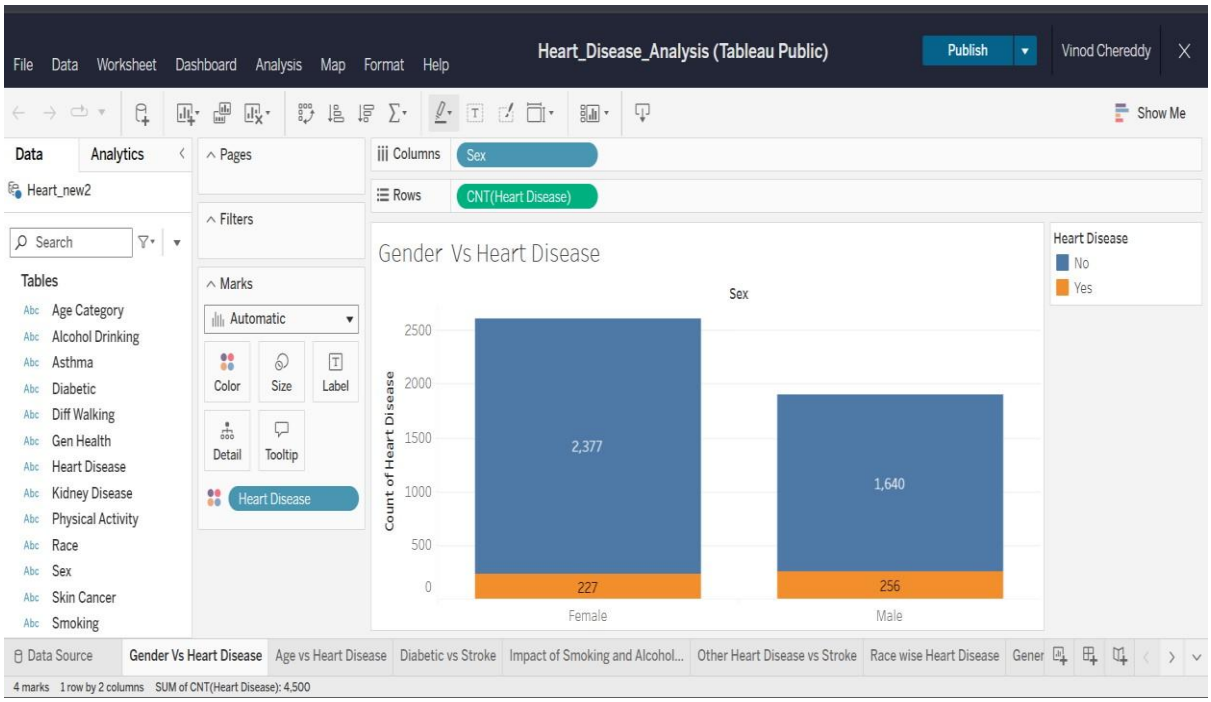
### 4.3 Data Visualization

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

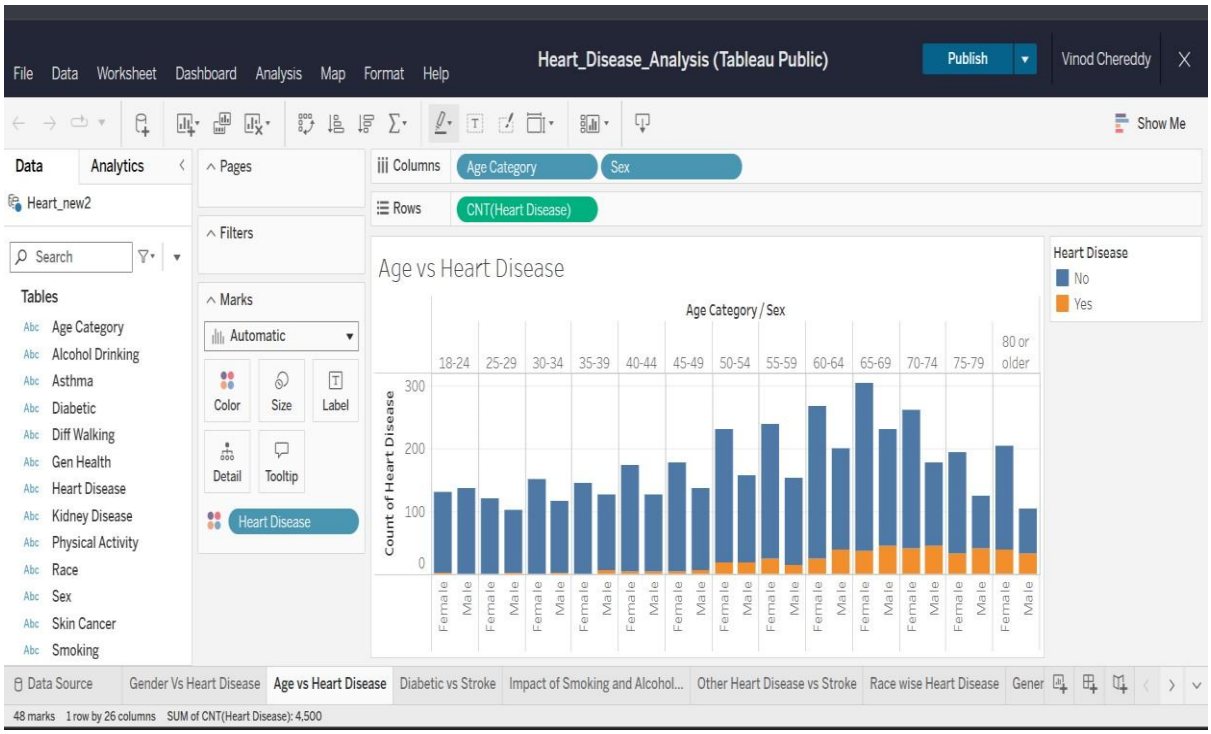
**No of Unique Visualizations**

This project focuses on analyzing heart disease data using Tableau by creating 8–10 unique visualizations including trend charts, heat maps, comparative bar graphs, scatter plots, and dashboards. The aim is to uncover risk factors like obesity, smoking, and lifestyle habits, enabling healthcare professionals, policymakers, and patients to gain actionable insights and make informed, preventive decisions against cardiovascular diseases.

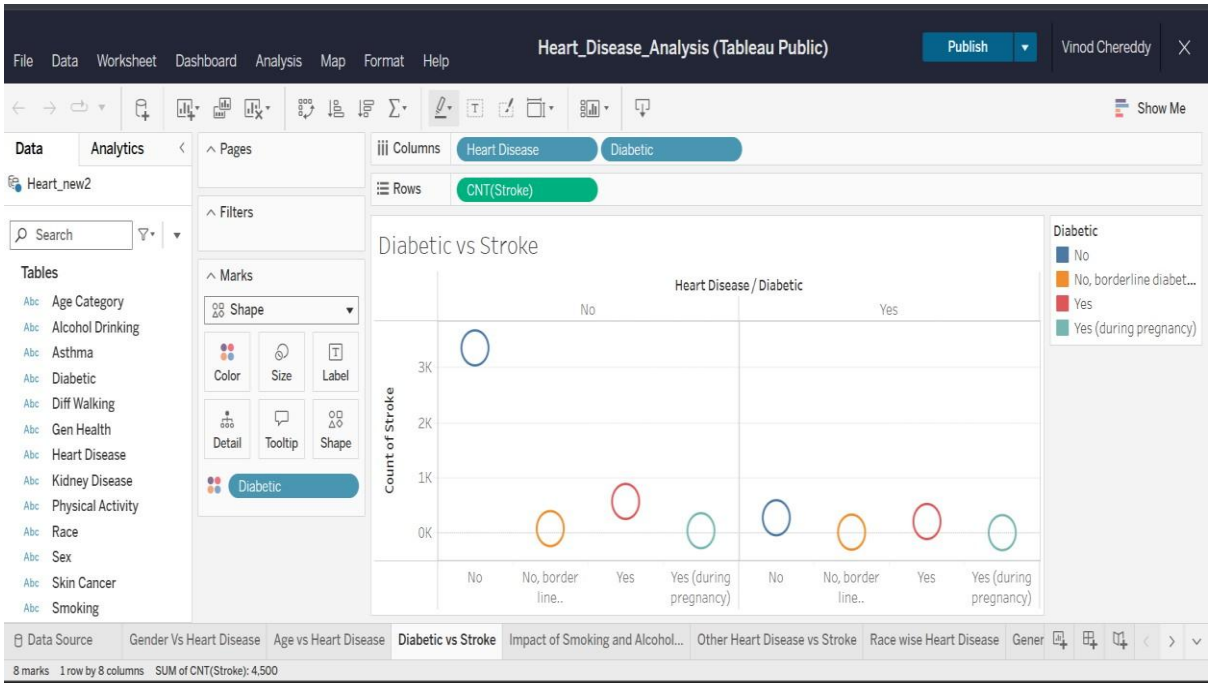
## 1. Gender  Vs Heart Disease



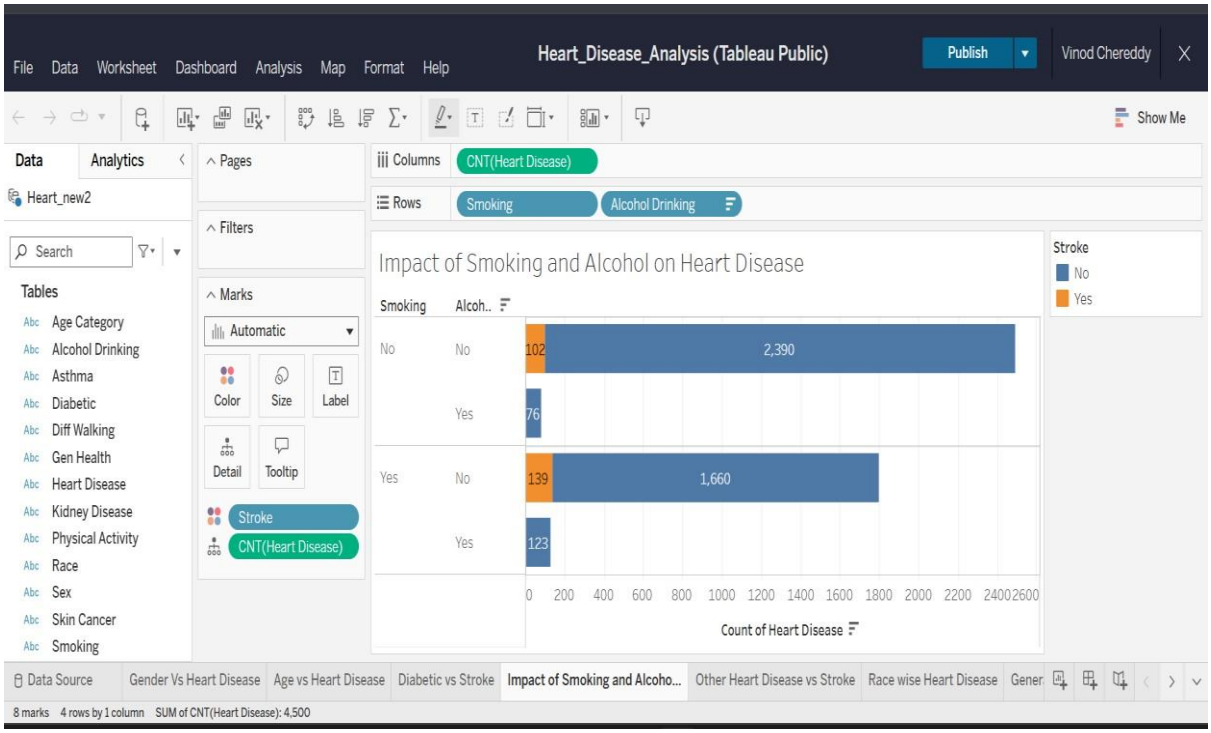

## 2. Age vs Heart Disease



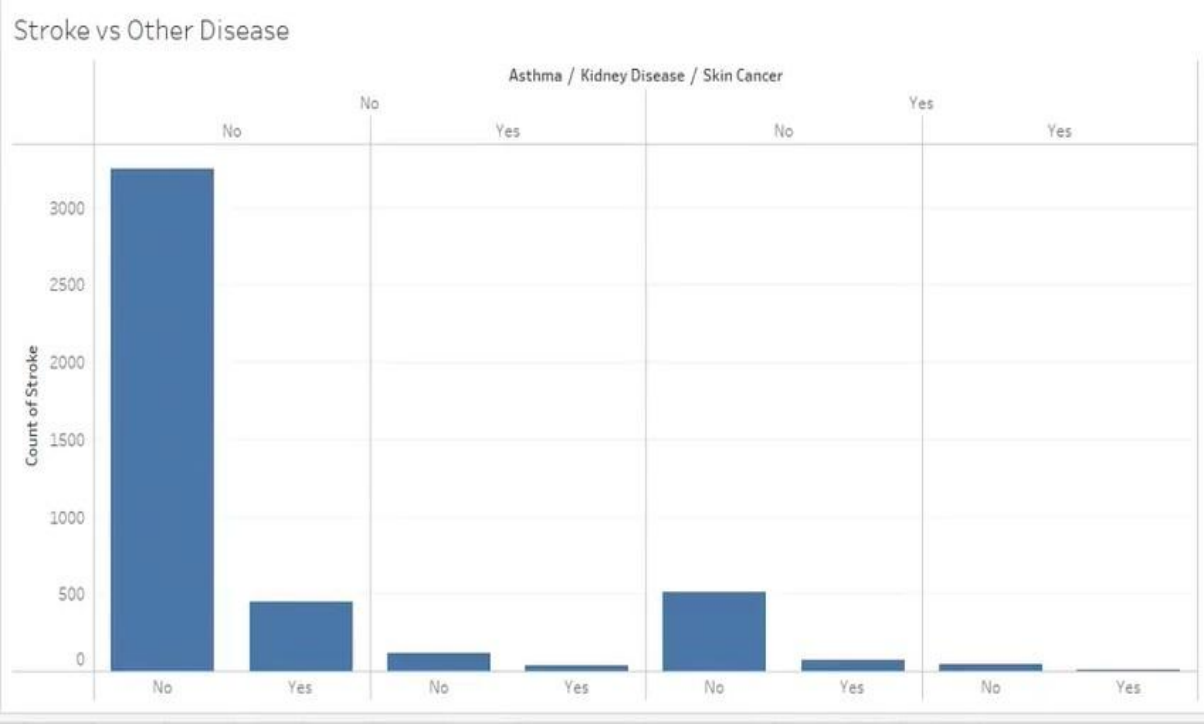
## 3.Diabetic vs Stroke



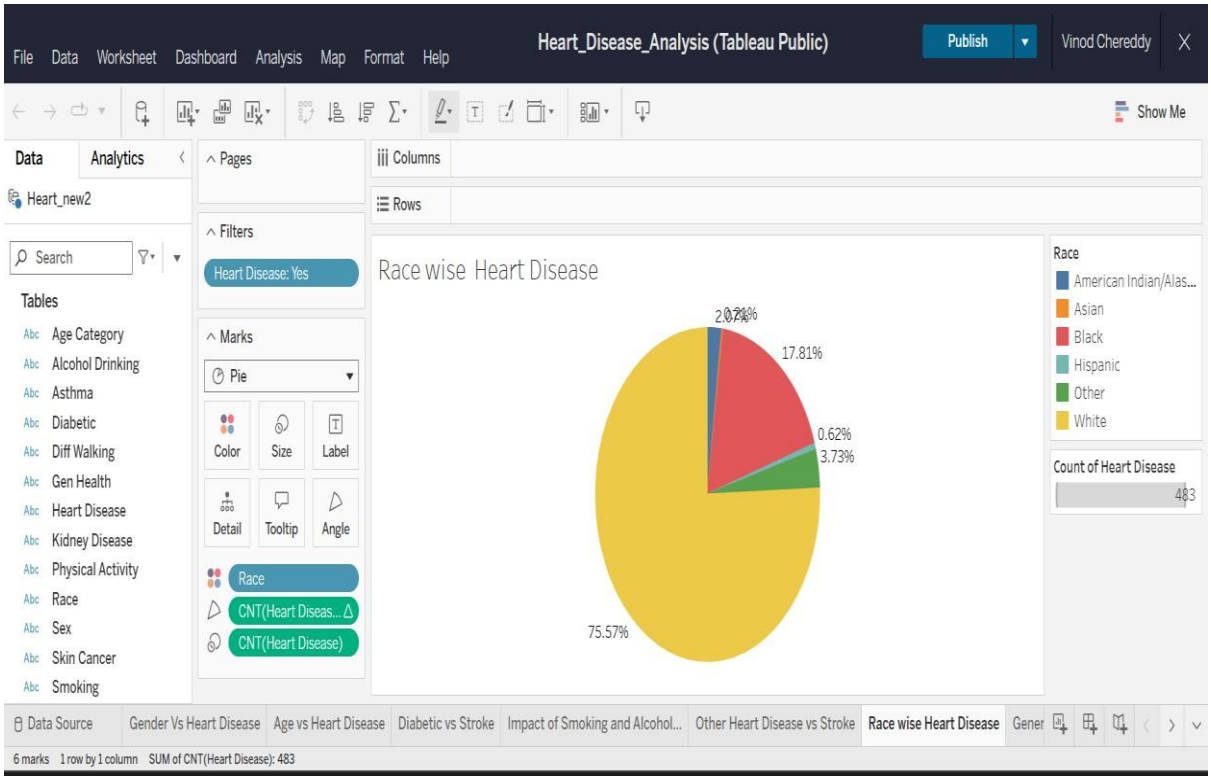

## 4. Impact of Smoking and Alcohol on Heart Disease



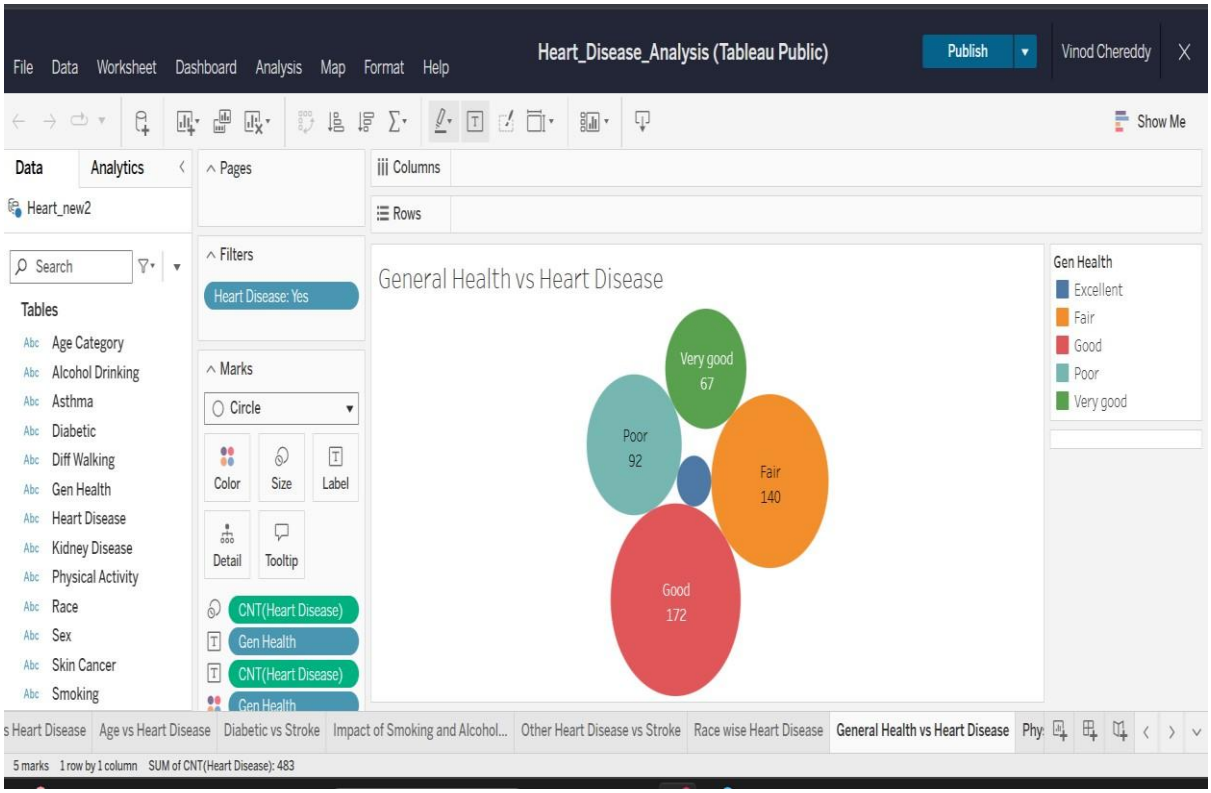
## 5. Other Heart Disease vs Stroke



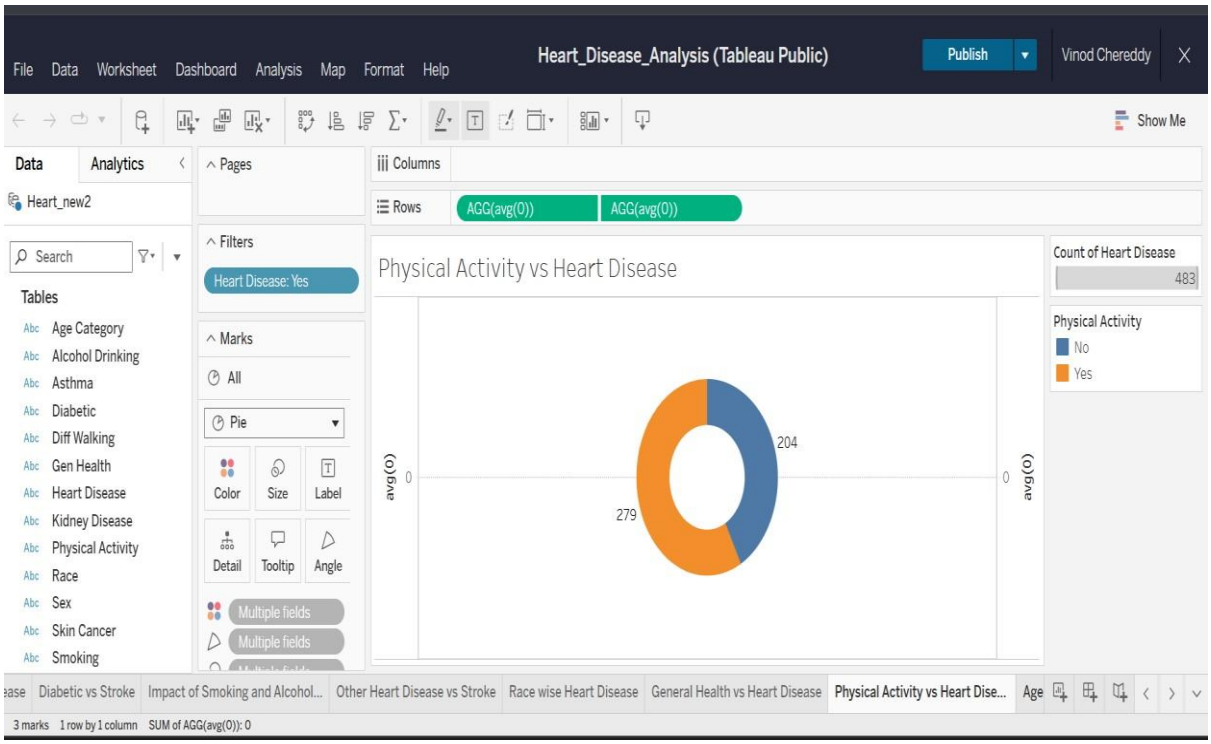

## 6. Race wise  Heart Disease



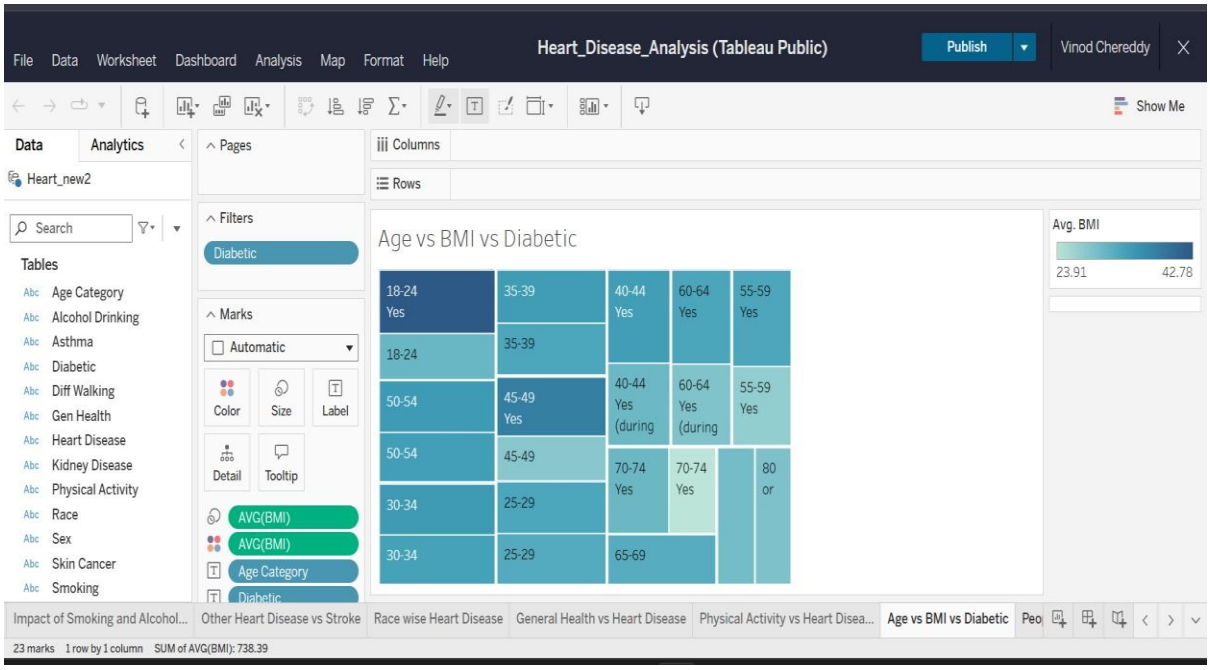
## 7. General Health vs Heart Disease



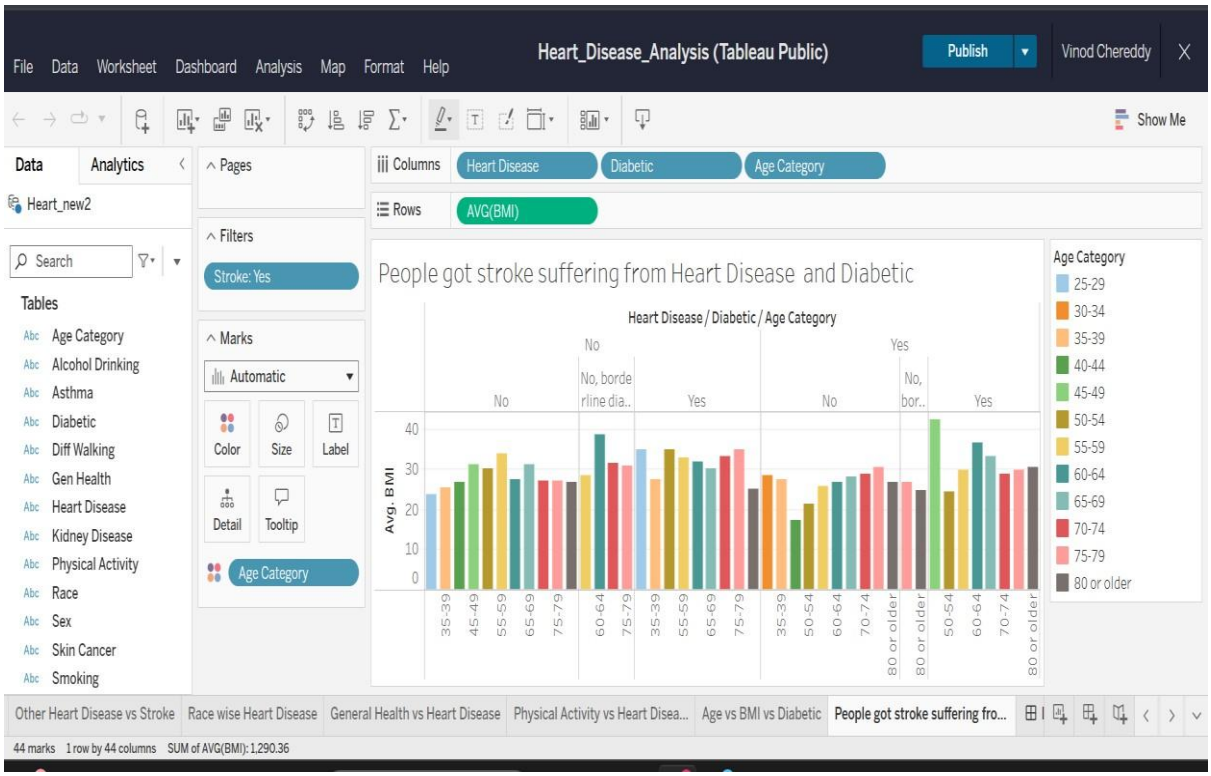

## 8. Physical Activity vs Heart Disease



## 9. Age vs BMI vs Diabetic




## 10. People got stroke suffering from Heart Disease and Diabetic




## 4.4 Proposed Solution

The proposed solution is a web-based heart disease analysis system that integrates data visualization and user interaction. The system uses **Tableau** as the primary visualization tool to create dashboards and stories that represent heart disease data in an intuitive and meaningful manner. These dashboards provide insights into various parameters such as age, gender, lifestyle habits, and associated medical conditions.

To enhance accessibility, the Tableau dashboards and stories are embedded into a **Flaskbased web application**. This enables users to access the system through a standard web browser without requiring any specialized software. The proposed solution focuses on ease of use, clarity of information, and responsiveness, ensuring that users can explore data smoothly.

The system supports interactive exploration, allowing users to apply filters, compare data points, and analyse trends in real time. By presenting data in a structured and visual format, the proposed solution helps users gain actionable insights that support awareness, early detection, and preventive healthcare planning.

### 4.3 Solution Architecture

The solution architecture defines the overall structure of the system and explains how different components interact to deliver the desire DJs content. The architecture is designed to ensure smooth data flow, efficient processing, and seamless user interaction.

The system follows a layered architectural approach consisting of three main components:

1. **User Layer:**
   This layer represents the end users who access the application through a web browser. Users interact with dashboards and stories to explore heart disease insights.

2. **Application Layer:**
   The Flask web application acts as the intermediate layer between users and the visualization system. It manages routing, user requests, and integration with Tableau dashboards.

3. **Data and Visualization Layer:**
   This layer consists of the heart disease dataset and the Tableau visualization engine. The dataset is processed, analysed, and converted into interactive visual representations.

Data flows from the dataset to the visualization layer, where it is transformed into dashboards and stories. These insights are then delivered to users through the web application. This modular architecture improves system scalability, maintainability, and performance.

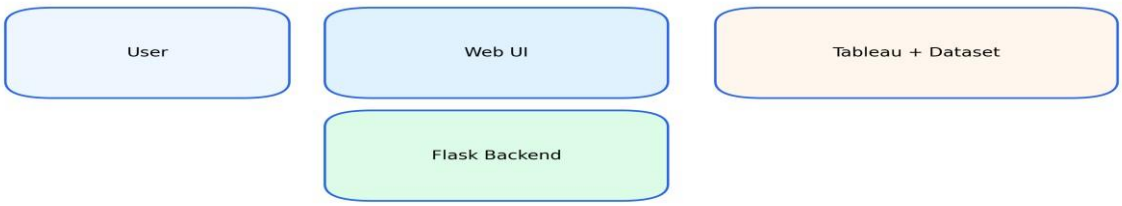



**Figure 17: Solution Architecture**

# 5. PROJECT PLANNING & SCHEDULING

Project planning and scheduling is an essential phase that defines how the project activities are organized, executed, and monitored over time. This phase ensures that the project is completed systematically within the available time and resources. Proper planning helps in reducing risks, improving coordination, and ensuring timely completion of all project milestones.

For the **Heart Disease Analysis** project, planning focused on dividing the overall work into manageable phases, assigning clear objectives to each phase, and ensuring a logical sequence of activities. This structured approach helped in maintaining consistency across ideation, analysis, design, development, and testing stages.

## 5.1 Project Planning

Project planning involved identifying key tasks, defining objectives for each task, and allocating time for successful execution. The planning process began with understanding the project scope and requirements, followed by organizing tasks in a step-by-step manner.

The project was planned across multiple phases:

- **Requirement Understanding:**
  Analysis of the problem domain, understanding heart disease data, and identifying user needs.
- **Design Phase:**
  Creation of problem–solution fit, proposed solution, and system architecture to define how the system will function.
- **Development Phase:**
  Preparation of datasets, creation of Tableau dashboards and stories, and integration with the Flask web application.
- **Testing Phase:**
  Verification of system functionality, performance testing, and validation of visual outputs.
- **Documentation & Demonstration:**
  Preparation of project documentation, screenshots, and demonstration planning.

Each phase was planned with clear goals to ensure smooth transition from one stage to the next. Time was allocated efficiently to balance data analysis, visualization, and documentation work. Regular progress tracking helped in identifying issues early and making necessary adjustments to the plan.

Effective project planning ensured that the system was developed in a structured manner, met all project objectives, and delivered meaningful insights within the scheduled timeline.

# 6. FUNCTIONAL AND PERFORMANCE TESTING

Testing is an important phase of the project lifecycle that ensures the system works as intended and meets the specified requirements. In this phase, both functional and performance aspects of the system are evaluated to verify accuracy, reliability, and efficiency.

For the Heart Disease Analysis project, testing was carried out to confirm that the dashboards, stories, and web integration function correctly and provide consistent results.

## 6.1 Performance Testing

Performance testing focuses on evaluating how efficiently the system responds under normal usage conditions. The main objective of performance testing in this project is to ensure that the application loads data smoothly, visualizations respond quickly to user interactions, and the system remains stable during operation.

Several performance parameters were considered during testing. These include dashboard loading time, response time when applying filters, and smooth rendering of charts and stories. The system was tested using different user interactions such as navigating between dashboards, applying multiple filters, and switching between story views.

The results of performance testing indicated that the Tableau dashboards loaded within an acceptable time frame and responded efficiently to user actions. The web application successfully handled data visualization without noticeable delays or performance degradation. These results confirm that the system is capable of providing a smooth and responsive user experience.

Overall, performance testing verified that the system meets the expected performance standards and is suitable for practical use in analyzing heart disease data and supporting decision-making.

# 7. RESULTS

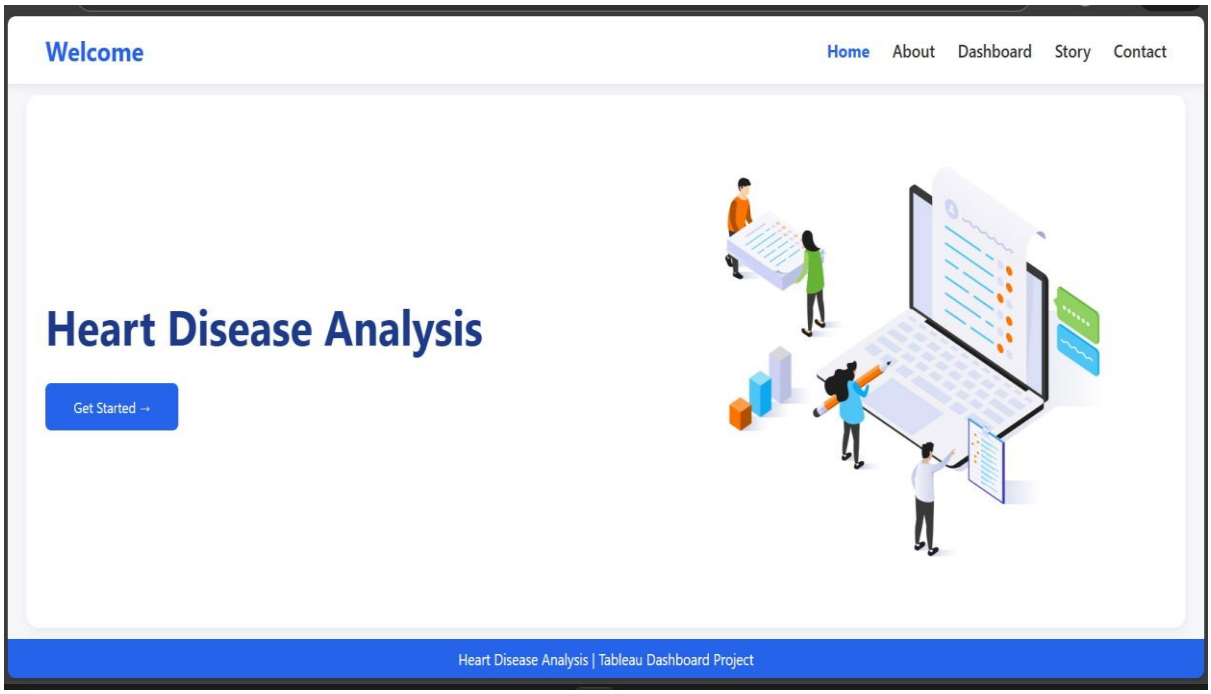
The results section presents the final outcomes of the **Heart Disease Analysis** project. This section demonstrates how the proposed system successfully transforms raw heart disease data into meaningful and interactive visual insights. The results validate the effectiveness of the system in supporting data-driven analysis, improving clarity, and enhancing user understanding of heart disease risk factors.

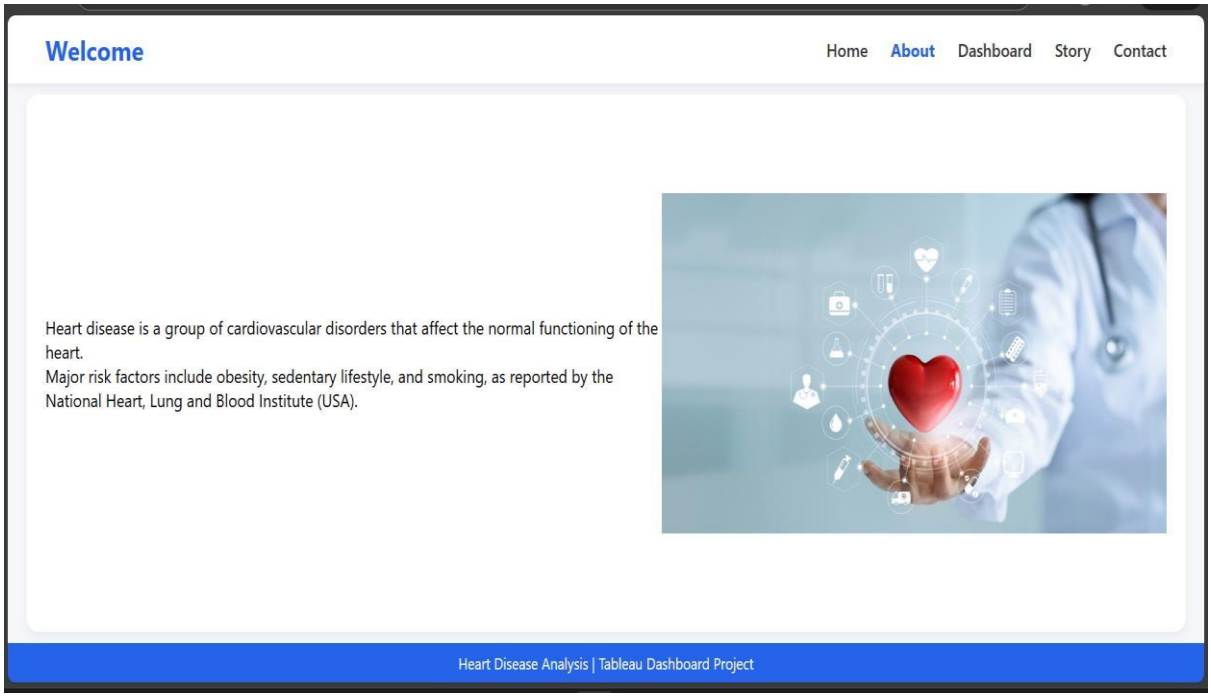
Through the use of Tableau dashboards and stories integrated with a web application, the system provides users with an intuitive way to explore trends, patterns, and correlations related to heart disease. The results confirm that the system meets its objectives of usability, responsiveness, and visual clarity.

## 7.1 Output Screenshots

### Dashboard and Story embed with UI With Flask

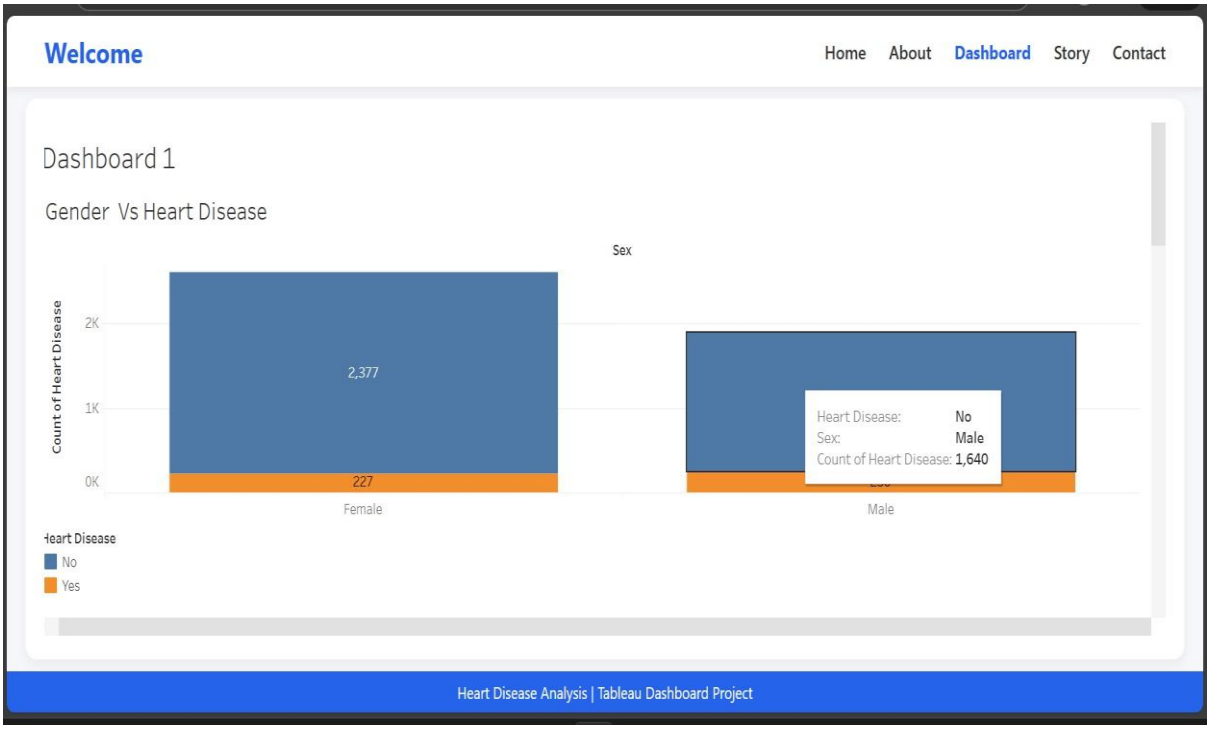
This section demonstrates how to embed Tableau Dashboards and Stories within a Flask web application to provide interactive, real-time access to heart disease insights. By integrating Tableau's powerful visualization capabilities with Flask's lightweight framework, users can explore dashboards directly in a browser, making the analysis more accessible, shareable, and actionable for healthcare professionals, policymakers, and individuals.





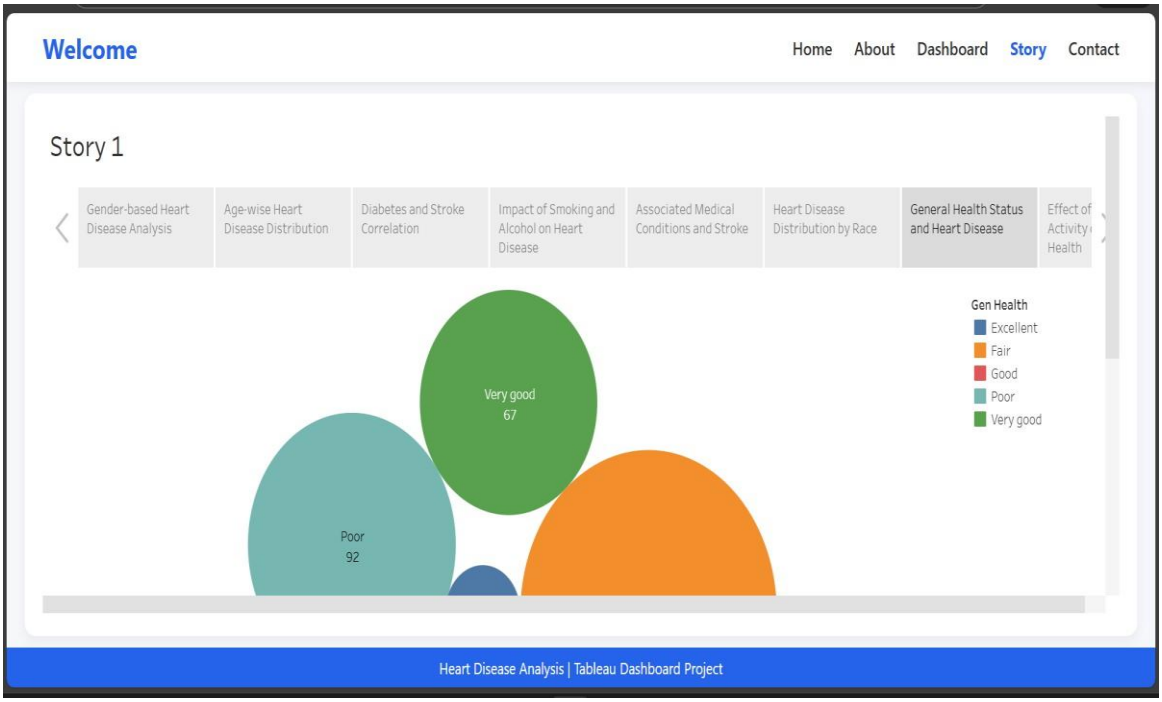

- **Dashboard View:**

  Displays key heart disease indicators such as age distribution, gender comparison, lifestyle impact, and associated medical conditions.




- **Story View:**

Presents a sequential narrative of insights, helping users understand trends and conclusions step by step.



# 8. Advantages

The proposed heart disease analysis system offers several benefits that improve healthcare data understanding and decision-making:

- **Easy Interpretation of Complex Data:**
  The system converts large and complex datasets into visual dashboards and stories, making the data easy to understand even for non-technical users.

- **Interactive Visualization:**
  Users can interact with dashboards by applying filters, comparing parameters, and exploring trends dynamically, which enhances analytical flexibility.

- **Improved Awareness and Preventive Insight:**
  Visual analysis helps identify key risk factors such as age, lifestyle habits, and medical conditions, supporting early awareness and preventive healthcare.

- **Web-Based Accessibility:**
  Integration with a web application allows users to access the system through a standard browser without installing specialized software.

- **Scalable and Extendable:**
  The system can be expanded in the future by adding more datasets, additional parameters, or advanced analytical techniques.

## 8.2 Disadvantages

Despite its advantages, the system also has certain limitations:

- **Dependence on Data Quality:**
  The accuracy of insights depends entirely on the quality and completeness of the dataset used.

- **Limited to Historical Data:**
  The current system analyzes historical data and does not include real-time data processing.

- **No Predictive Analysis:**
  The project focuses on descriptive analysis and visualization and does not include predictive or machine learning models.

- **Internet Requirement:**
  Access to the system requires a stable internet connection to load dashboards and visualizations.

# 9. CONCLUSION

The **Heart Disease Analysis** project successfully demonstrates how data visualization and web-based technologies can be effectively used to analyze and present complex healthcare data. By transforming raw heart disease datasets into interactive dashboards and stories, the project makes it easier to understand important patterns, trends, and risk factors associated with heart disease.

The system addresses the challenges of traditional data analysis by providing a clear and user-friendly visual approach. Users are able to explore data dynamically, apply filters, and gain meaningful insights without requiring advanced technical knowledge. The integration of Tableau dashboards within a Flask-based web application further enhances accessibility and usability.

The project outcomes confirm that visual analytics plays a significant role in improving awareness, supporting preventive healthcare decisions, and enabling data-driven analysis. Overall, the Heart Disease Analysis project meets its objectives and provides a practical solution for analyzing heart disease data in an efficient and understandable manner.

# 10. APPENDIX

This section provides supplementary information related to the project, including source code references, dataset details, and external links used during development.

**Source Code (if any)**

The source code for the project includes:

- Python Flask application for web integration
- HTML and CSS files for the user interface
- Configuration files for embedding Tableau dashboards

(Source code files are maintained locally and can be shared upon request.)

**Dataset Link**

The dataset used in this project contains heart disease–related information, including demographic details, lifestyle factors, and medical conditions.

**Dataset:** Heart Disease Dataset (CSV format)

Dataset used for analysis and visualization in Tableau.

**GitHub & Project Demo Link** •

**GitHub Repository:**

**github-Heart-Disease-Analysis**

- **Project Demo / Tableau Public Link:**

  **public.tableau Heart_Disease_Analysis**